D4.1 Natural selection

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d4-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 790–797 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

GENE POOLS AND SELECTION

D4.1.9 Gene pool

p. 790

A **gene pool** is all the genes and alleles in an interbreeding population. **Allele frequency** is the proportion of all copies of a gene that are one allele. Evolution is a change in allele frequencies over generations.

D4.1.10 Separated populations

pp. 791–792

Separated populations differ in mutations, selection pressures and chance events. In humans, lactase persistence is common in northern Europe but rare in East Asia, while an ALDH2 variant, which causes alcohol flush, is common in East Asia.

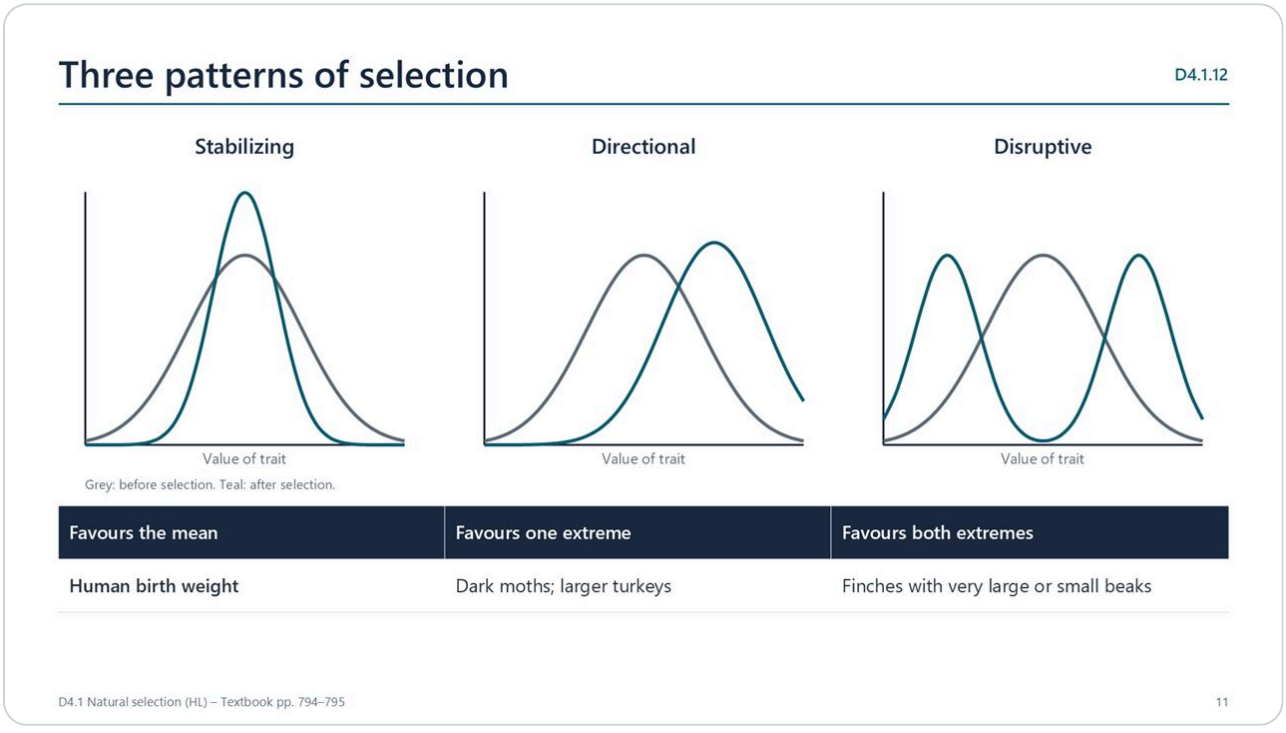
D4.1.11 Selection and allele frequency

pp. 793–794

Individuals with favourable heritable traits survive and reproduce more, so they pass on more copies of their alleles and those alleles rise in frequency.

D4.1.12 Three patterns of selection

pp. 794–795



Stabilizing selection favours intermediate phenotypes, **directional** favours one extreme and **disruptive** favours both extremes.

HARDY-WEINBERG AND ARTIFICIAL SELECTION

D4.1.13 Hardy-Weinberg equation

pp. 796–797

$p + q = 1$ and $p^2 + 2pq + q^2 = 1$. Start from the homozygous recessive frequency: q is its square root, then $p = 1 - q$, and carriers are $2pq$.

D4.1.14 Conditions for equilibrium

p. 797

Random mating, no selection, no mutation, no migration and a very large population. If observed frequencies do not fit, at least one condition is not being met.

D4.1.15 Artificial selection

p. 797

People choose which individuals breed, to increase wanted traits in crops and domestic animals. Antibiotic resistance is an unintended result of human action, but it is natural selection.

© 2026 Akash Parthi Bendran. Please ask before using or sharing: see whatthecell.org/about.html#permission